

1978 and the Passing Revolution (1978–1999)

Two rule changes in one spring, and the twenty-year arms race they started.

Put on a game from the early 1970s. The picture looks familiar: two wide receivers, a tight end, two backs, a quarterback under center. Then watch a receiver run a route. A linebacker hits him as he leaves the line, a safety hits him again eight yards downfield, and nobody throws a flag, because none of it is illegal. Watch an offensive tackle pass-block: his fists are balled against his chest and his elbows are out, like a man shielding himself in a crowd. The league trimmed that downfield contact twice in the mid-1970s and it made little difference: in 1977 NFL teams averaged **17.2 points a game**, the fewest since 1942. Twenty-one years later a single team, the 1998 Minnesota Vikings, scored 556 points, a league record at the time. What happened in between? The answer starts with two short changes to the rulebook in March 1978, and with the twenty years of move and counter-move they set off, a story in which every offensive idea you now see on Sundays met a defensive answer, and every answer started the next round.

You'll be able to:

- State the two 1978 rule changes (the Mel Blount rule and open-hand pass blocking), say what each one changed on the field, and read the jump they produced in the league's passing numbers.
- Tell Air Coryell from the West Coast offense by what each asks the quarterback to do, and recognize Gibbs's one-back offense, the run-and-shoot and the K-Gun on old film.
- Explain each defensive answer as a response to something specific: nickel and dime to the third receiver, Arnsperger's 53 and the 3-4 to the need to disguise a fourth rusher, the 46 to the two-back offense, Lawrence Taylor to slow pass protection, the zone blitz to the quick passing game.
- Explain why the left tackle became the blind-side tackle, and what an offense does about the best pass rusher on the field.
- Explain what the 1994 rule package changed (the two-point conversion and kickoffs from the 30), and do the expected-points arithmetic behind the two-point decision.
- Explain free agency (1993) and the salary cap (1994), and name three ways they changed how teams build and coach.

i You'll need

This is a history chapter, so it leans on almost everything before it. The four direct prerequisites, one line each:

- **From the Single Wing to the 1970s Scoring Drought:** by the mid-1970s defenses had the upper hand. Defenders could hit receivers downfield (the chuck rule limited that only in 1974 and 1977), and scoring fell to its lowest level in decades, which this book calls the 1970s scoring drought.
- **Offensive Systems and Play-Calling Languages:** an offensive system is a language plus a core menu of plays. You met Air Coryell and its digit system, the West Coast offense, Erhardt-Perkins and the run-and-shoot there as *languages*; this chapter tells where they came from and what they were answering.
- **Zone Blitzes, Fire Zones, and Simulated Pressure:** a zone blitz rushes a linebacker or defensive back, drops a lineman into coverage and plays zone behind it; the fire zone (five rush, three under, three deep) is its most common form. That chapter tells the Arnsperger–LeBeau origin story in full.
- **Even Fronts:** four-man fronts, technique numbers, and Buddy Ryan's 46 defense, whose bear front covers the center and both guards.

And a few terms from elsewhere that come up constantly here: nickel (five defensive backs) and dime (six), from A First Look at Defense; the 3-4 and the stand-up edge, from Odd and Hybrid Fronts; personnel groupings such as 12 personnel (one back, two tight ends), from Personnel Groupings; the pro set (two backs side by side) and the singleback formation, from Classic Formations; counter, from Gap Schemes; and the pass-protection words punch, kick slide and chip, from Pass Protection.

Reading the diagrams: offense is blue circles (Q quarterback; R or T the running back, F the fullback, H a halfback or H-back; X, Y, Z the receivers, Y usually the tight end; on the line, T G C G T are the tackles, guards and center). Defense is outlined squares (E end, T tackle, N nose, L or B linebackers, with M, W and S for the Mike, Will and Sam; CB corners; FS and SS safeties). Solid arrows are routes and runs, dashed orange arrows are defenders' drops and options, a bar at the end of a line is a block, and the dashed brown line is the pass. A green ring marks the player to watch. The left-edge numbers give depth in yards past the line of scrimmage.

Where the story starts: the 1977 problem

The previous chapter ended with the defenses on top. By the mid-1970s the best of them, Pittsburgh's Steel Curtain above all, had found that the cheapest way to stop a pass was to stop the receiver from running his route. Until 1974 linebackers and safeties were allowed to hit a receiver anywhere on the field until the ball was in the air, so a tight end crossing the middle could be knocked off his path two or three times before the quarterback was ready to throw. On the other side of the ball, pass blockers had to keep their hands in to their chests. They blocked with forearms, elbows and shoulders, and a quick defensive end could get past a man who wasn't allowed to reach out and grab hold of him.

The league had already tried small fixes. In 1974 it limited downfield contact on receivers (the "Isaac Curtis rule," after a Cincinnati receiver whose speed defenses could only answer by hitting him), moved kickoffs back from the 40 to the 35 and the goalposts from the goal line to the end line, all "to add action and tempo." In 1977 it allowed defenders to contact a receiver only once, and banned the head slap that pass rushers like Deacon Jones had used to stun blockers.¹ None of it worked. In 1977 teams scored 17.2 points a game; only 40 percent of offensive plays were passes; and 5.7 percent of passes were intercepted, more than one in eighteen.

The league had reasons to care beyond taste. In October 1977 Pete Rozelle signed television contracts with all three networks that would carry every regular-season game from 1978, then the largest television package ever negotiated, and an expanded 16-game schedule was coming with them.² A sport selling itself to a national audience every week needed points, and points in the 1970s came mostly from long, grinding drives that defenses were winning more often every season.

¹Pro Football Hall of Fame, Chronology of Football, 1974 ("sweeping rules changes were adopted to add action and tempo to games": kickoffs moved from the 40 to the 35, goalposts moved to the end lines, "the extent of downfield contact a defender could have with an eligible receiver was restricted"): <https://www.profootballhof.com/football-history/football-history/1960-1979/1974> ; 1977 ("Rules changes were adopted to open up the passing game and to cut down on injuries. Defenders were permitted to make contact with eligible receivers only once; the head slap was outlawed"): <https://www.profootballhof.com/football-history/football-history/1960-1979/1977> . The 1974 restriction is often called the "Isaac Curtis rule": "Isaac Curtis," Wikipedia, https://en.wikipedia.org/wiki/Isaac_Curtis . The 1977 averages quoted here (17.2 points per team-game, 40.0% of offensive plays passes, 5.7 interceptions per 100 attempts) are computed from Pro Football Reference's league averages; see the note on the chart.

²Pro Football Hall of Fame, Chronology of Football, 1977 (October 12: contracts with the three networks to televise all regular-season and postseason games for four years beginning in 1978, "the largest single television package ever negotiated"; March 29: a 16-game regular season adopted to begin in 1978): <https://www.profootballhof.com/football-history/football-history/1960-1979/1977>

Spring 1978: two rules, one direction

In March 1978 the owners passed a package of changes. Two of them did almost all the work.³

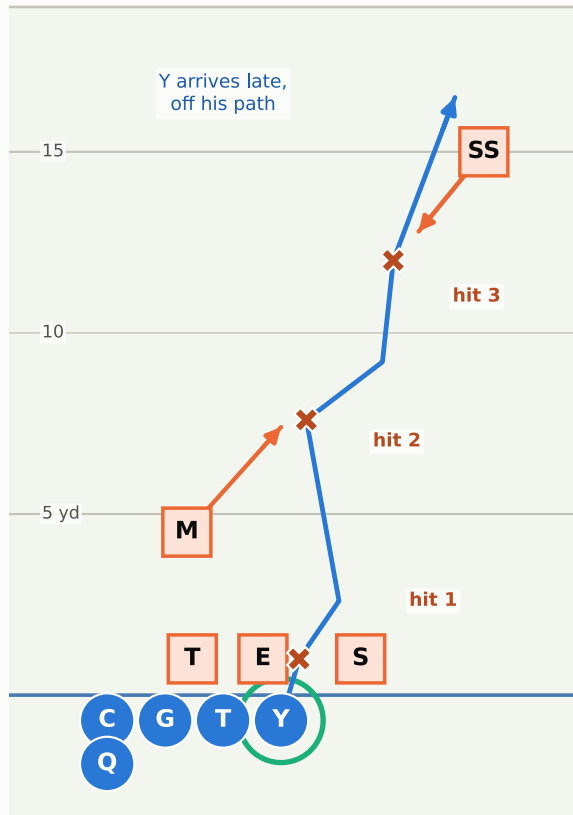
The receiver goes free: the Mel Blount rule

The first rule drew a line five yards past the line of scrimmage. Inside that line, a defender may still contact a receiver in front of him, jamming him, rerouting him, knocking him off the line. Beyond it, while the quarterback still has the ball, he may not initiate contact that impedes the receiver: no shove, no chuck, no collision. Once the ball is thrown, any contact that interferes is pass interference, as before. That five-yard zone is the **Mel Blount rule (1978)**: *a defender may contact a receiver only within five yards of the line, before the throw*. The name honors, if that is the word, Pittsburgh's 6-foot-3 cornerback Mel Blount, the 1975 defensive player of the year, whose bump-and-run coverage was so physical that the league's answer was named after him. Some accounts attach the nickname to the 1977 one-chuck rule instead; the five-yard rule of 1978 is the one most people mean.⁴

³Pro Football Hall of Fame, Chronology of Football, 1978 ("The NFL continued a trend toward opening up the game. Rules changes permitted a defender to maintain contact with a receiver within five yards of the line of scrimmage, but restricted contact beyond that point. The pass-blocking rule was interpreted to permit the extending of arms and open hands," March 17; a seventh official, the side judge, added March 14): <https://www.profootballhof.com/football-history/football-history/1960-1979/1978>. "1978 NFL season," Wikipedia (contact "only to a point of five yards beyond the line of scrimmage... usually referred to as the 'Mel Blount Rule'"; "pass blocking rules were extended to permit extended arms and open hands"; the side judge added "to help rule on legalities downfield"; 16-game schedule; a second wild card): https://en.wikipedia.org/wiki/1978_NFL_season. Today's rule: NFL Rulebook, Rule 8, Section 4: <https://operations.nfl.com/the-rules/nfl-rulebook/>

⁴"Mel Blount," Wikipedia (6 ft 3 in; AP defensive player of the year 1975 with a league-leading 11 interceptions; "the rules regarding pass coverage were changed making such harassment of a receiver illegal. The rule would come to be named the Mel Blount Rule"): https://en.wikipedia.org/wiki/Mel_Blount; "Mel Blount explains play that created Mel Blount rule," Steelers Depot (Feb. 2024): <https://steelersdepot.com/2024/02/mel-blount-explains-play-that-created-mel-blount-rule>. On the nickname's attachment to the 1977 change in some accounts, see Christopher L. Gasper, "How wide receivers have evolved in the NFL," *Boston Globe* (Sept. 4, 2013): <https://www.bostonglobe.com/sports/2013/09/04/how-wide-receivers-have-evolved-nfl/akAnTGNJaS7OrVnjAbbN6j/story.html>

Up to 1973: hit him anywhere, as often as you like



From 1978: contact only in the first 5 yards

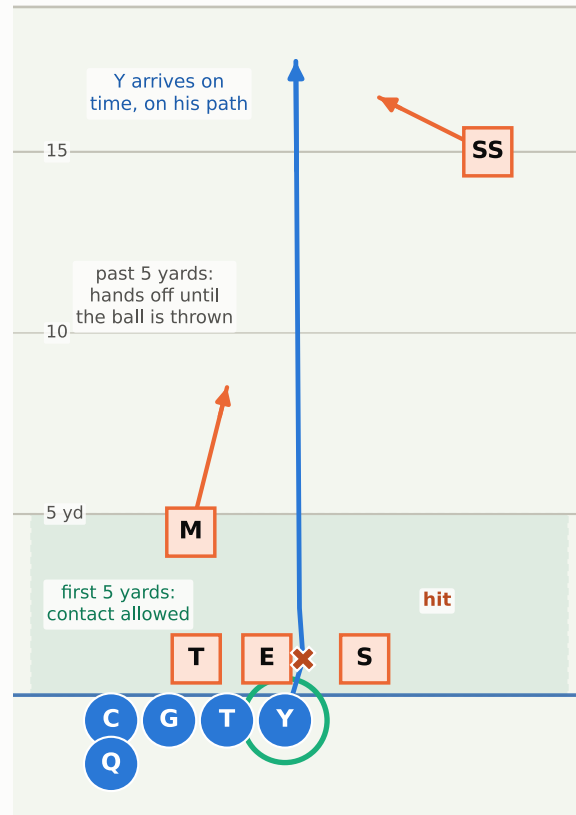


Figure 1: The 1978 contact rule, on one tight end's seam route (illustrative). Left, the rules before 1974: the Sam (S) hits Y at the line, the Mike (M) knocks him off his path at 8 yards, and the strong safety hits him again at 12, so he arrives late and wide of where his quarterback expects him. (In 1974 the league limited downfield contact, and in 1977 it allowed only one hit.) Right, from 1978: contact is legal only in the green zone, the first five yards past the line. Past it the Mike and the safety may not touch Y until the ball is thrown, so he runs his seam on time and on his path.

Figure 1 shows what changed for the defense. Before, every defender between the receiver and his landmark was a potential speed bump. A zone defense could *re-route* receivers: knock crossers off their paths, steer a tight end toward a waiting safety, and destroy the timing a passing game depends on. After 1978, only the defenders near the line got a shot, and only once the receiver was past them was he free. A linebacker dropping to 8 yards could no longer hit the tight end running past him. He had to *cover* him, with his feet, and linebackers built to take on guards are not built to run with tight ends.

The deepest effect was on **timing**. A route that can be disrupted anywhere can't be thrown on a count, because the quarterback never knows when his receiver will arrive. A route that can be disrupted only in the first five yards can be. If the receiver gets a clean release, he will be at 12 yards, breaking toward the sideline, at the same moment on Sunday that he was in Wednesday's practice. The quarterback can throw to the *spot* before the receiver turns around. Every timing route you have met in this book, from the quick game to the dropback concepts, assumes that.

The blocker gets his hands: open-hand pass protection

The second rule was about the other end of the play. From 1978 an offensive lineman pass-blocking could **extend his arms and open his hands**. That sounds like a technicality; it changed the geometry of the pocket. A blocker restricted to his chest and forearms meets the rusher body to body, and a rusher who gets his shoulder past the blocker's hip is past him. A blocker who can extend his arms meets the rusher at arm's length, three feet away, and can *steer* him: feel which way he is going and push him further that way, past the quarterback, while staying between him and the passer. That is the punch and the kick slide of Pass Protection, and neither exists without the 1978 rule. Holding (grabbing and restricting the rusher) stayed illegal. What changed is that a lineman could now use his hands at all.

More time in the pocket matters most to the throws that take the longest. A three-step drop and a quick slant need little protection; a seven-step drop and a 20-yard post need a lot. So the second rule pointed the same way as the first: toward the deep and intermediate passing game the 1970s defenses had strangled.

! Why it exists

The two rules attacked the 1970s defensive advantage from both ends at once. The Blount rule took away the defense's best way to wreck a route (hitting the receiver), and the hands rule took away its best way to wreck the quarterback's clock (beating a blocker who couldn't use his arms). The league's stated aim, in the Hall of Fame's words, was to "open up the game."⁵ What the rules gave up was a style of defense. A team built on big, physical defensive backs who won by contact, like the Steelers who gave the rule its name, now had to win with speed and position instead.

And the rest of the package

Three smaller changes rode along, and each fits the same story. The schedule grew from 14 games to 16 and a second wild card was added in each conference, which meant more games to televise. The league added a **seventh official**, the side judge, partly to rule on what happens downfield: a new contact rule needs someone standing where the contact happens. And the next spring, in 1979, officials were told to blow the whistle as soon as a quarterback was "clearly in the grasp" of a tackler, the first of many rules protecting the passer that the league's investment in passing would require.⁶

🔥 Common misconception

"The Mel Blount rule outlawed bump-and-run coverage." It didn't. Inside five yards a cornerback can still jam a receiver at the line, and teams have played press coverage ever since, from the 1980s Raiders to Seattle's press Cover 3 of the 2010s. What the rule removed was the *second*, third and fourth hit downfield, the ones that let a zone defense steer receivers wherever it wanted. It was also not the last word: the same rule went loosely enforced for a quarter of a century until officials were told to call it strictly in 2004, the "Ty Law rule" of Coverage Fundamentals.

i Madden vs. real life

In a video game the rulebook is fixed, and the best plays are the best plays forever. In the NFL the rulebook is rewritten every spring, and schemes chase the rewrites. The 1978 changes are the clearest example in the sport's history: no coach invented a play that ended the 1970s scoring drought. The owners did, by changing what a defender was allowed to do with his hands, and then the coaches spent twenty years finding out what that made possible. When a rule changes, ask the question this chapter asks over and over: *what did this take away, and from whom?*

What the numbers did

How big was the change? Season-level league totals answer that. Pro Football Reference keeps a year-by-year table of league averages, and Figure 2 draws four of its rates for every season from 1970 to 1999.⁷ Rates matter here, not totals, because the schedule grew from 14 to 16 games in 1978 and two seasons (1982 and 1987) were cut short by strikes.

⁵Pro Football Hall of Fame, Chronology of Football, 1978, cited above ("The NFL continued a trend toward opening up the game"). Pittsburgh, the team Blount played for, won the next two Super Bowls by throwing more: Terry Bradshaw was the 1978 league MVP and threw for 318 yards and four touchdowns in the 35–31 win over Dallas in Super Bowl XIII.

⁶Pro Football Hall of Fame, Chronology of Football, 1979 (rules changes "instructed officials to quickly whistle a play dead when a quarterback was clearly in the grasp of a tackler," March 16): <https://www.profootballhof.com/football-history/football-history/1960-1979/1979> ; for 1978's 16 games and second wild card, see the 1978 note above.

⁷Pro Football Reference, "NFL League Year-by-Year Averages" (per team per game and season totals), <https://www.pro-football-reference.com/years/NFL/index.htm> , as archived by the Internet Archive on Dec. 28, 2024: <https://web.archive.org/web/20241228065223/https://www.pro-football-reference.com/years/NFL/index.htm> . Points per game and net yards per attempt are PFR's per-game figures; passes as a share of plays is pass attempts divided by pass attempts plus rushing attempts (sacks are not included, so the true share of dropbacks is a little higher); interceptions per 100 passes is interceptions divided by attempts. Seasons are regular seasons; 1982 (nine games) and 1987 (fifteen, three with replacement players) were shortened by strikes (Pro Football Hall of Fame, Chronology of Football, 1982 and 1987: <https://www.profootballhof.com/football-history/football-history/1980-1999/1987>). Pro Football Reference blocks automated access; spot-check the figures there before print.

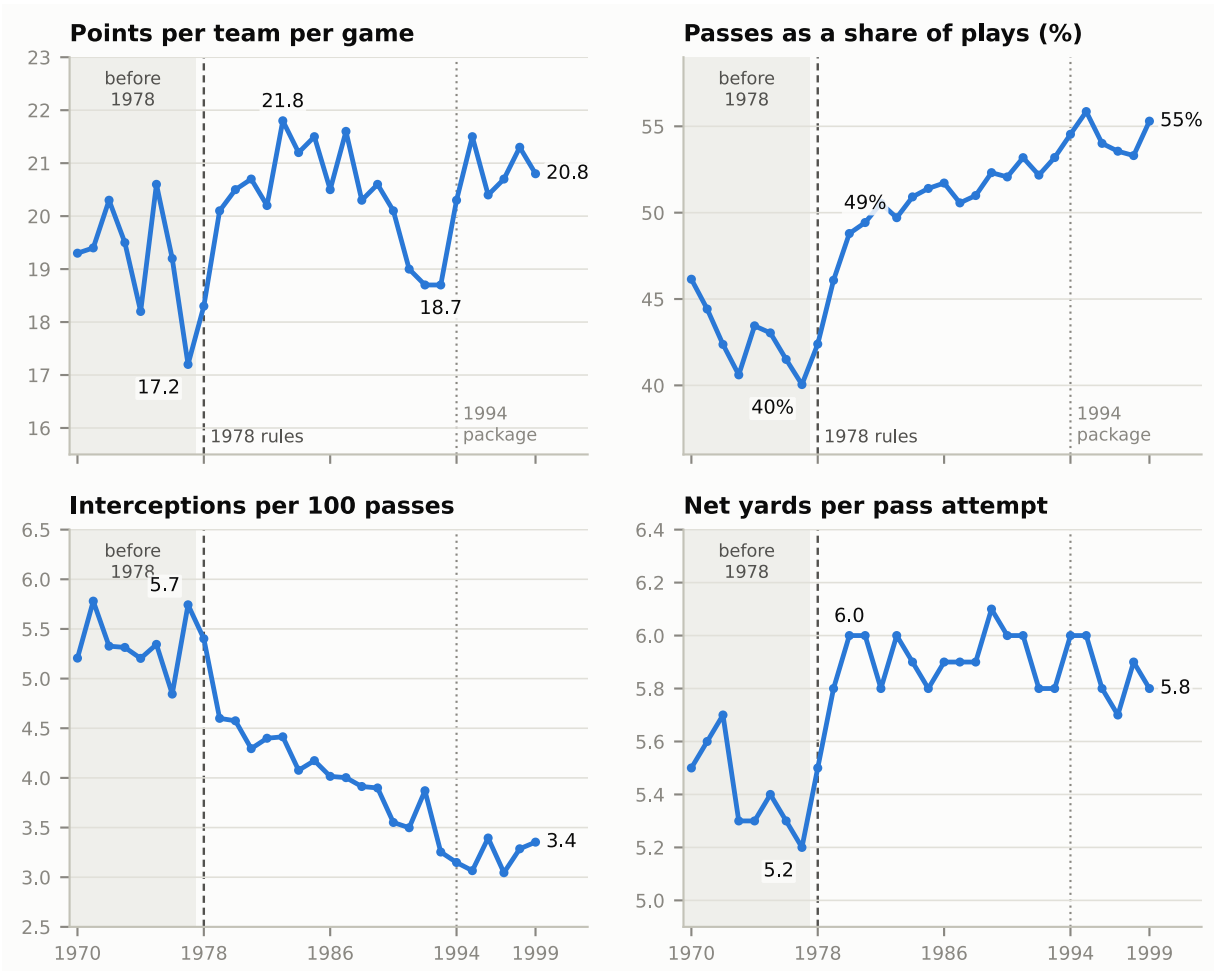


Figure 2: The NFL's passing game, 1970 to 1999: four league-wide rates by season, from Pro Football Reference's league averages. The shaded years are before the 1978 rules (dashed line); the dotted line marks the 1994 rule package. Passes as a share of plays jumped from 40% in 1977 to 49% in 1980 and kept rising. Interceptions per 100 passes fell from 5.7 to 3.4 over twenty years. Net yards per pass attempt rose from 5.2 to 6.0 and then stopped. Scoring rose from 17.2 points to the low 20s, sagged in the early 1990s and rose again after 1994.

Read Figure 2 panel by panel.

- **Passes as a share of plays** (top right) is the cleanest signal, and the biggest. In 1977 teams threw on 40 percent of their plays. By 1980 it was 49 percent, and after 1983 it never went below half again. Offenses decided, almost at once, that throwing had become a better deal.
- **Interceptions** (bottom left) fell from 5.7 per 100 passes in 1977 to 4.6 in 1979 and 1980, and kept falling for twenty years, to 3.4 in 1999. A receiver who can't be knocked off his route is where the quarterback expects him to be, and a ball thrown to the right spot is harder to intercept. Completion rate rose too, from 51 percent in 1977 to 56 percent in 1980.
- **Net yards per pass attempt** (bottom right: passing yards minus sack yardage, divided by attempts plus sacks) jumped from 5.2 in 1977 to 6.0 in 1980 and then stopped. Through the 1980s and 1990s it wandered between 5.7 and 6.1. Each throw got a little better, once, and then the defenses caught up with it.
- **Points** (top left) went from 17.2 to 20.5 in three seasons. Note 1978 itself: only 18.3. The rules arrived that season, but the offenses that exploited them took a year or two to arrive.

The pattern in those last two panels is the theme of this chapter. A rule change opens a gap, the offense pours into it, and then the defense adapts and the efficiency gain levels off. What stays is the *volume*: teams kept throwing more, because once the defense adjusted to stop the pass, it had less left over to stop the run. And at the far right of the points panel is the next round of the same cycle: scoring sagged to 18.7 in 1992 and 1993, and the league changed the rules again (the 1994 package, later in this chapter).

The coach most ready for 1978 was already in the league. **Don Coryell** had built pass-first offenses at San Diego State in the 1960s and with the St. Louis Cardinals from 1973, and early in 1978 he took over the San Diego Chargers. Coryell's offense, built on Sid Gillman's idea that a defense should be made to cover every blade of grass, put receivers in motion, lined them up all over the formation, and made the deep throw the first option on most passing plays. His quarterback, **Dan Fouts**, had spent five unremarkable seasons in San Diego. In 1979 he threw for 4,082 yards, breaking Joe Namath's single-season record of 4,007; in 1981 he threw for 4,802, and he became the first quarterback to pass for 4,000 yards in three straight seasons. The 1979 Chargers were the first AFC West champion to call more passes than runs, and San Diego led the league in passing yards every season from 1978 to 1983.⁸

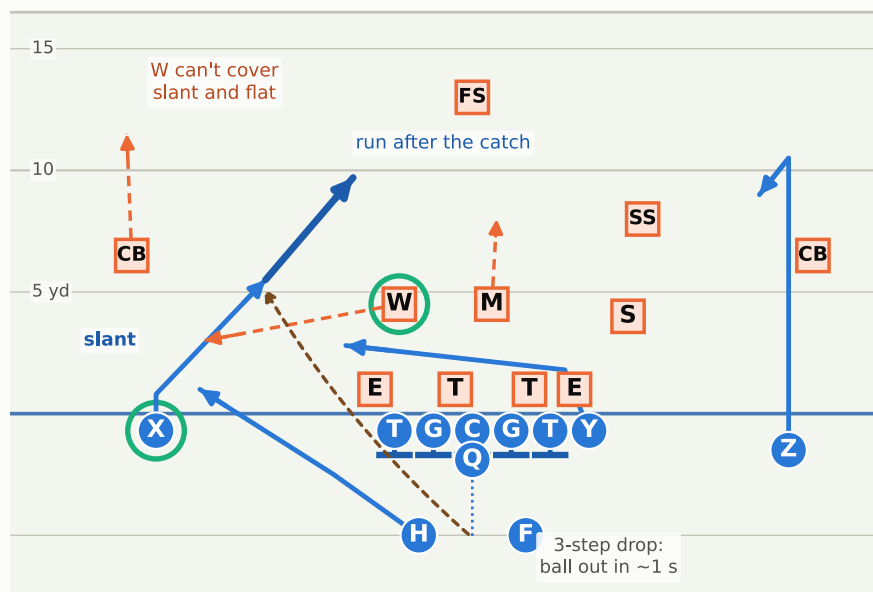
Diagram illustrating a 7-step drop drill for a 100m sprint. The diagram shows a starting block (X) and a finish line (Z). A solid blue line represents the 7-step drop path, with a dashed brown line representing the 7-step drop path. Key points include: X (start), Y (7-step drop), Z (finish), and various intermediate points (A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z). The diagram also shows a 'seven block: line + both backs' and a '7-step drop'.

Figure 3 is a Coryell-style call in the digit system of Offensive Systems: “896” gives X an 8 (a post), Y a 9 (a go, here up the seam), and Z a 6 (a dig). Three receivers attack three spots deep, and against the two deep safeties of Cover 2 the tight end’s seam splits them. Notice what pays for it: both backs stay in to block, seven blockers for a seven-step drop, so the quarterback has time to let the deep routes develop.

⁸⁴“Air Coryell,” Wikipedia (the Chargers led the league in passing yards 1978–1983 and in 1985; “The Chargers in 1979 were the first AFC Western Division champion to run more passing plays (541) than rushing (481)”); motion; “Throwing a deep pass was the first option on each play”): https://en.wikipedia.org/wiki/Air_Coryell; “Dan Fouts,” Wikipedia (4,082 yards in 1979, breaking Joe Namath’s record of 4,007; 4,802 in 1981; first to throw for 4,000 yards in three straight seasons): https://en.wikipedia.org/wiki/Dan_Fouts; Pro Football Hall of Fame, Don Coryell: <https://www.profootballhof.com/players/don-coryell/>. Coryell’s lineage through Zampese, Turner and Martz: Chris B. Brown, cited in Offensive Systems.

game always costs: long drops, longer protection, fewer receivers in routes, and interceptions when the throw is late. Coryell's Chargers had famously poor defenses and never reached a Super Bowl. His ideas lasted anyway. They ran through his assistants (Joe Gibbs, Ernie Zampese) to Norv Turner's Dallas offenses of the 1990s and to Mike Martz's 1999 St. Louis Rams, the "Greatest Show on Turf."

Offense's second answer: Walsh stretches it sideways



⁹“The Epic in Miami,” Wikipedia (January 2, 1982; San Diego led 24–0 at the end of the first quarter; Chargers 41, Dolphins 38 in overtime; Fouts 33 of 53 for 433 yards, NFL postseason records; Winslow 13 receptions, a playoff record, for 166 yards and a touchdown, plus a blocked field goal; NFL 100 Greatest Games ranked it fourth): https://en.wikipedia.org/wiki/Epic_in_Miami

Walsh's offense treats the short pass as a run. Figure 4 is the idea in its plainest form: a three-step drop, and the ball out in about a second to X on a quick slant, while the halfback runs to the flat behind him and the tight end crosses underneath. The Will linebacker can't cover both the slant and the flat, a horizontal stretch, so whichever he chooses, the quarterback throws the other. The receiver catches the ball in stride, already running, and the yards come after the catch.

Why the 1978 rules made it work. Every route in Figure 4 crosses a defender's face within a few yards of the line, exactly where re-routing used to wreck them. After 1978, once the receiver was five yards downfield, the linebacker in his path could do nothing but try to beat him to the spot, and on a timed throw he never will. Walsh's offense is the timing game at its purest: drops matched to route depths, the ball thrown to the spot before the receiver turns, and a progression the quarterback can run in a second and a half. It needs less protection than Coryell's, throws fewer interceptions, and turns a 5-yard completion into a 15-yard gain. The cost is that it needs an extremely accurate quarterback and receivers who run every route to the inch, and that it is long and hard to learn: the West Coast play calls you met in Offensive Systems spell out every receiver's job because the timing has no slack.

Walsh's 49ers won Super Bowls after the 1981, 1984 and 1988 seasons, and his assistants carried the system everywhere (the coaching tree is in Case Study: Bill Walsh's 49ers and the West Coast Offense). Between them, Coryell and Walsh defined the two ways to pass after 1978: stretch the defense vertically, or stretch it horizontally. Nearly every passing game since mixes the two.

Defense's first answer: more defensive backs

Defenses answered the new passing game first with **personnel**. If offenses were going to throw on half their plays, and to put a third receiver on the field to do it, then a linebacker covering that third receiver was a losing matchup on half the snaps. The fix was to take a linebacker off and put a fifth defensive back on: the nickel. The idea was older than 1978 (it is usually traced to Philadelphia assistant Jerry Williams around 1960, named by George Allen, and popularized by Don Shula and Bill Arnsparger's Miami Dolphins in the 1970s),¹¹ but after 1978 it went from a special package to an everyday one. In long-yardage situations defenses went further, to the dime: six defensive backs.

Each substitution is a trade. A nickel defense has one fewer player built to take on blockers, so it is softer against the run. And substitution became a game of its own: offenses changed their personnel to draw a defense into the wrong package, the substitution matching of Personnel Groupings. A defense in nickel against two tight ends is too light; a defense in base against three receivers is too slow. Hold on to that trade; it decides the K-Gun story later in this chapter.

Defense's second answer: Arnsparger's 53 and the 3-4

The second answer was about the *front*, and it had started in Miami before 1978. **Bill Arnsparger**, Don Shula's defensive coordinator, asked the Dolphins to trade for a Cleveland linebacker named **Bob Matheson** before the 1971 season, and built a package around him. Matheson lined up at the end of the line on the tight end's side, standing up. On some snaps he rushed the passer like a defensive end; on others he dropped into coverage like a linebacker. The package was named for his jersey number: the **53 defense**, *Arnsparger's front in which one hybrid player decided, snap by snap, whether the defense was a 4-3 or a 3-4*. The 1972 Dolphins, Arnsparger's "No-Name Defense," went 17-0.¹²

¹¹"Nickel defense," Wikipedia (Jerry Williams, Philadelphia, 1960; George Allen and the name; "popularized by head coach Don Shula and defensive coordinator Bill Arnsparger of the Miami Dolphins in the 1970s"): https://en.wikipedia.org/wiki/Nickel_defense ; "Dime defense," Wikipedia: https://en.wikipedia.org/wiki/Dime_defense . The usual account is discussed in A First Look at Defense.

¹²"Bob Matheson," Wikipedia (acquired before the 1971 season at Arnsparger's request; "Matheson's versatility against both the run and pass was at the heart of the '53' defense, a scheme that was named for his jersey number and devised by Arnsparger. Depending on whether the situation was running or passing, he was a strong-side defensive end and a fourth linebacker respectively"): https://en.wikipedia.org/wiki/Bob_Matheson ; "Bill Arnsparger," Wikipedia (the No-Name Defense; the 1980s Killer B's): https://en.wikipedia.org/wiki/Bill_Arnsparger . Arnsparger's "rush five guys and cover with six" and "safe pressure": Chris B. Brown, "Controlled Chaos," Grantland (Sept. 26, 2012): <https://grantland.com/features/dick-lebeau-evolution-coverage-tactics-zone-blitz/>

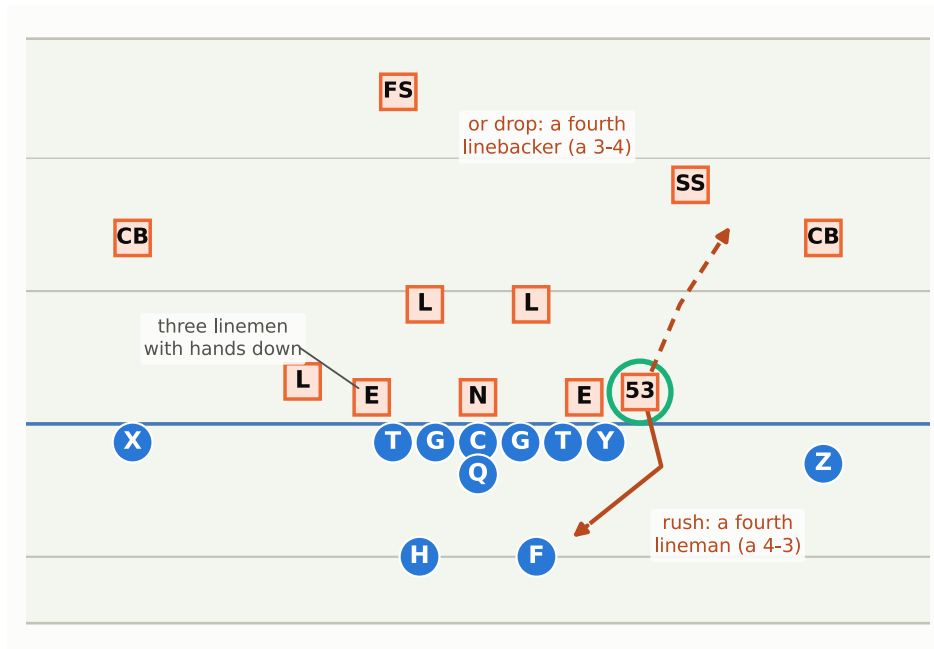


Figure 5: Arnsperger's 53 (illustrative, against a 1970s pro set). Three linemen have a hand down; Bob Matheson, number 53 (ringed), stands at the end of the line outside the tight end. At the snap he either rushes, making a four-man line (a 4-3 look), or drops into coverage, making four linebackers (a 3-4 look). The blockers have to decide before the snap whether to account for him, and they can't know which he will do.

Look at Figure 5 from the quarterback's side. Before the snap there are three linemen with a hand on the ground and a fourth man standing at the end of the line. Is it four rushers or three? The quarterback and his blockers have to decide where Matheson is going before he goes, and they have no way to know. If they block him and he drops, a blocker is wasted and a linebacker is covering a receiver they thought was open. If they ignore him and he rushes, he is a free rusher. Arnsperger's own phrase for what the 53 gave him, in a later interview, was the ability to "rush five guys and cover with six" from a look that showed neither. He called it "safe pressure," and it is the seed of the zone blitz.

The rest of the league took the alignment more than the trick. Through the mid-1970s, teams that had learned the Oklahoma 5-2 in college stood up its ends and called it a 3-4 (the competing origin claims are laid out in Odd and Hybrid Fronts), and by the late 1970s and early 1980s it was the league's most common base defense. The 3-4 fitted the post-1978 game for a reason that had nothing to do with the nose tackle: its outside linebackers stand up at the end of the line, where they can rush or drop, so a 3-4 is a 53 on every snap. The offense never knows for certain which of the two edge players is coming. And it creates a place on the field for a particular kind of athlete, a linebacker who is really a pass rusher, the stand-up edge. In 1981 the most famous one of all arrived.

Lawrence Taylor and the blind side

The New York Giants took **Lawrence Taylor** from North Carolina with the second pick of the 1981 draft and played him at outside linebacker in their 3-4. He was the league's defensive player of the year as a rookie, again in 1982, and a third time in 1986, when he had 20.5 sacks and became the first defensive player voted the league's MVP unanimously. Bill Parcells, his defensive coordinator and then head coach, and Bill Belichick, the Giants' defensive coordinator from 1985, built the defense around the fact that no one could block him.¹³

Taylor's speed broke the way offenses protected the passer. Through the 1970s, a blitzing linebacker was a running back's problem: the back stayed in to block whichever linebacker came. A running back blocking Lawrence Taylor was a mismatch, and teams spent the 1980s inventing better answers. San Francisco's Walsh, in a January 1982 playoff game against the Giants, assigned a guard to him, which other teams copied; then coaches moved the job to the tackle, and in the end to whole protection schemes built around where Taylor was (the slide, the chip and the help side of Pass Protection).

¹³"Lawrence Taylor," Wikipedia (second overall pick in 1981; Defensive Rookie of the Year 1981; Defensive Player of the Year 1981, 1982, 1986; 20.5 sacks and the first unanimous MVP by a defensive player in 1986; Walsh assigned a guard to him "in the 1982 playoffs," later tackles; Gibbs developed the two-tight-end offense and the H-back in response): https://en.wikipedia.org/wiki/Lawrence_Taylor ; Pro Football Hall of Fame, Lawrence Taylor: <https://www.profootballhof.com/players/lawrence-taylor/>

Where Taylor lined up mattered as much as how fast he was. He rushed most often from the defense's right side, the offense's left, and that is where a right-handed quarterback is most exposed.

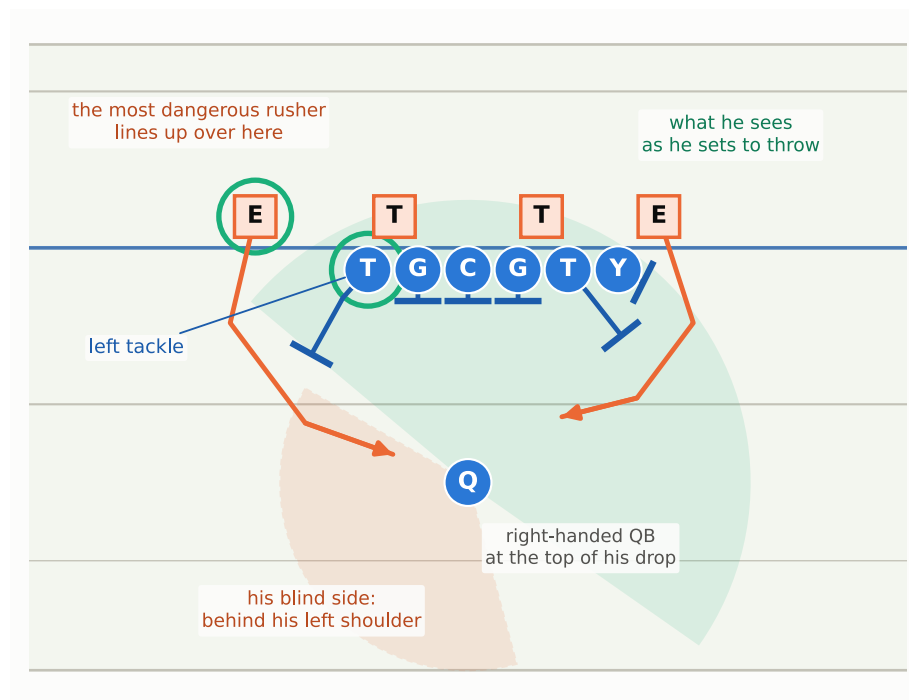


Figure 6: Why the left tackle matters (illustrative). A right-handed quarterback at the top of his drop. As he sets to throw, his left shoulder points downfield and his eyes go with it, so he can see the area shaded green but not the area shaded orange, behind his left shoulder. The edge rusher on the right (the defense's left) comes at him where he can see and avoid him. The one on the left (ringed) comes from the blind side, and if he beats the left tackle (ringed) around the edge, he arrives unseen. Defenses put their best pass rusher on that side, so offenses put their best pass blocker there to meet him.

Figure 6 shows why. A right-handed quarterback dropping back from under center turns his body to the right as he drops, and when he sets to throw, his left shoulder points at his target and his eyes go downfield. Everything coming from behind his left shoulder is outside his field of view. A rusher from the right side is in front of him or at the edge of his vision, and he can step up or slide away from it. A rusher from the left arrives unseen, at full speed, at a quarterback holding the ball away from his body. That produces fumbles and injuries. Lawrence Taylor's sack of Washington's Joe Theismann on *Monday Night Football* in November 1985, which broke Theismann's leg and ended his career, came from that side (the scene opens Michael Lewis's book *The Blind Side*; see The Twenty-Two).

The result was a new idea of the offensive line. The **blind-side tackle** is *the tackle who protects the side the quarterback can't see: the left tackle for a right-handed passer*. Teams began to put their best pass blocker there, to draft him early and, once a salary cap made every dollar a choice (later in this chapter), to pay him like it. Two caveats keep the idea honest. For a left-handed quarterback (Steve Young in San Francisco, Mark Brunell in Jacksonville in the 1990s) the blind side is the right. And the side matters less once defenses learned to move their best rusher around, as they all do now, and once quarterbacks played from the shotgun and turned less on the drop; today the right tackle is paid almost as well. The blind side still pulls: when a team has to choose, its best pass protector usually plays on the left.

Gibbs: one back, two tight ends, and the counter Trey

Taylor's division rival had the most direct answer. **Joe Gibbs**, Coryell's former offensive coordinator in San Diego, became Washington's head coach in 1981 and played the Giants twice a year. Gibbs's solution, in his own words, was a game plan built for one man: "We had to try in some way have a special game plan just for Lawrence Taylor... he's one person that we learned the lesson the hard way. We lost ball games."¹⁴ He took the second back out of the backfield and put him beside the tackle as an extra blocker. Playbooks of the time labelled the backs Q, H and F, so the new player kept his letter: the **H-back**, a tight end or fullback who lines up a step off the line, outside the tackle, where he can block an edge rusher, chip him and release into a route,

¹⁴"H-back," Wikipedia (the name from the Q, H and F backs of Gibbs's notation; the position "was created by Gibbs as a direct response to Lawrence Taylor," with Gibbs's quotation): <https://en.wikipedia.org/wiki/H-back>. Gibbs's three titles (Super Bowls XVII, XXII and XXVI): Pro Football Hall of Fame, Chronology of Football, 1980–1999: <https://www.profootballhof.com/football-history/football-history/1980-1999/1992>

or go in motion to the other side. With a second tight end on the line, the formation became the one-back, two-tight-end set (12 personnel) that Washington made its base: the ace of Classic Formations.

The one-back formation answered more than Taylor. With one back, the defense couldn't key a fullback to find the run. With two tight ends and an H-back, the offense had extra blockers at the line who could also go out for passes, so the same formation threatened both. And it fitted the run game Washington built behind its huge offensive line, the "Hogs": the counter Trey, the guard-and-tackle counter of Gap Schemes, in which the back steps one way and two backside linemen pull the other way to lead him. The language of those runs, and Washington's "70 Chip" on fourth-and-1 in Super Bowl XVII, are in *The Language of the Run Game*.

Gibbs won three Super Bowls in ten seasons with three different starting quarterbacks (Joe Theismann, Doug Williams and Mark Rypien), which says something about how much of the offense was the system.

Buddy Ryan's 46, and how the one-back attacked it

While Gibbs was building his offense, Chicago's defensive coordinator **Buddy Ryan** was building the defense meant to destroy offenses like it. The 46 put eight men near the line, covered the center and both guards with a nose and two 3-techniques (the bear front), stacked both outside linebackers on the tight end's side, and usually played man coverage behind a blitz. It was designed against the two-back offenses of the early 1980s, and against them it was devastating: the 1985 Bears allowed the fewest points in the league and beat New England 46–10 in Super Bowl XX (the full plate and Ryan's story are in *Even Fronts*).¹⁵

The 46's strength was its aggression and its alignment rules: both outside linebackers go to the tight end. Against a one-back team with two tight ends, the rule becomes a question: *which* tight end? Whichever side the defense chooses, the other side has fewer defenders, and a counter run pulls two extra blockers to exactly that side.

Follow Figure 7. The defense sets its stack to the in-line tight end, Y, on the right: two outside linebackers (W and S) out there, plus the Mike over the right tackle. On the left the offense faces only the end, a tackle, the nose and the strong safety (SS), who in the 46 plays like a linebacker. The H-back lines up on that side. At the snap the back jabs right, toward the stack, and the Mike, reading him, steps that way too. Then the right guard and right tackle pull left. Count the left side now: the guard kicks out the end, the left tackle and left guard block down on the tackle and the nose, the H-back takes the safety, and the pulling tackle leads through the hole onto the Mike. Five blockers for five defenders, and the back has a hole. The two linebackers stacked on the tight end, the players who make the 46 so dangerous against plays toward them, are chasing from the wrong side of the field.

Notice what the offense gives up to do this. The center has to block back on the 3-technique the right guard leaves, the hardest block on the play, and the play only works if the defense commits to one side. The 46 had answers of its own: it could set its stack to the H-back, or flip it when he went in motion, or blitz the side the counter was headed for. That is the pattern of this whole era. The offense found a formation that forced the defense to declare, and the defense found a way to declare late.

¹⁵"46 defense," Wikipedia (Buddy Ryan; named for safety Doug Plank's jersey number; "most effective against the two back, two wide receiver sets that many teams commonly used in the 1980s"): https://en.wikipedia.org/wiki/46_defense ; Pro Football Hall of Fame, Chronology of Football, 1986 (Chicago 46, New England 10 in Super Bowl XX): <https://www.profootballhof.com/football-history/football-history/1980-1999/1986>

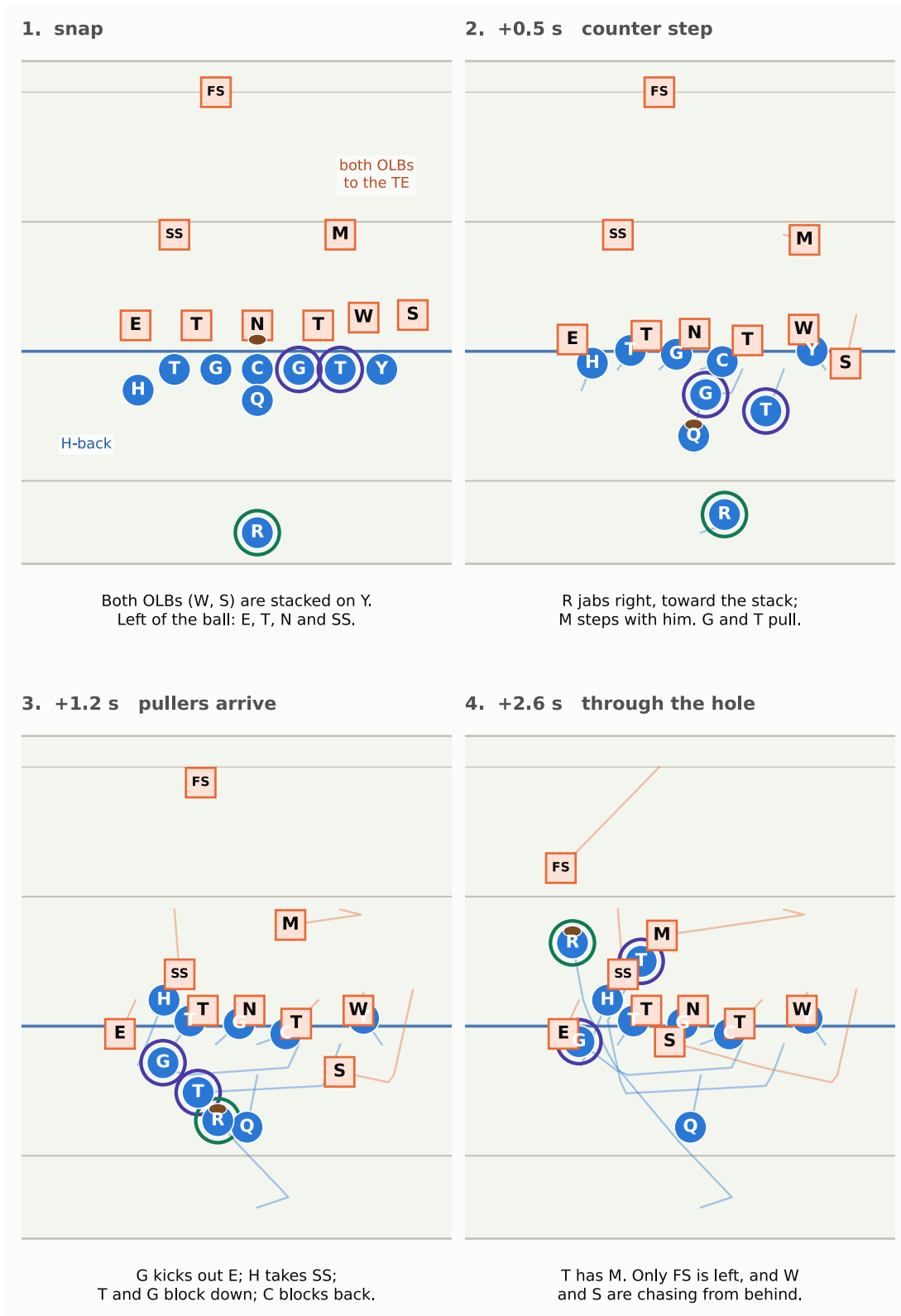


Figure 7: The counter Trey from a one-back, two-tight-end set against the 46 (simulated, BDB-format tracking; illustrative, not a specific play). The 46 sets both outside linebackers (W, S) and the Mike to the in-line tight end, Y, on the right. 1: left of the ball are only the end, a 3-technique, the nose and the strong safety, with the H-back lined up across from them. 2: the back (green ring) jabs right toward the stack and the Mike steps with him, while the right guard and right tackle (purple rings) pull left. 3: the guard kicks out the end, the left tackle and guard block down, the H-back takes the safety, and the center blocks back on the 3-technique the guard left. 4: the pulling tackle leads onto the Mike; the back is through the hole with only the free safety ahead and the stacked linebackers chasing from the far side. Faint lines are each player's path so far.

The game: Washington 42, Denver 10, at San Diego on January 31, 1988. Denver led 10–0 after the first quarter. Washington then scored 35 points in the second quarter, still the Super Bowl record: Doug Williams, the first Black quarterback to start and win a Super Bowl, threw four touchdown passes in that quarter alone and finished with 340 yards, and rookie Timmy Smith, making his first NFL start, ran for a Super Bowl record 204 yards.¹⁶

What to notice: the one-back sets with two tight ends, the H-back moving before the snap, and Smith's long runs, which come behind pulling linemen after a step the wrong way: the counter Trey. Then watch Ricky Sanders's two touchdown catches in the same quarter, of 80 and 50 yards, and ask yourself whether you could have told run from pass before the snap. That uncertainty is the point of the formation.

The rivalry with Ryan went on after he left Chicago to coach the Philadelphia Eagles (1986–1990). In November 1990 his Eagles beat Washington 28–14 on *Monday Night Football* in the “Body Bag Game,” knocking nine Washington players out, both quarterbacks among them; seven weeks later, in the wild-card round in Philadelphia, Washington won 20–6, and Ryan was soon out of a job.¹⁷

Spreading the field: the run-and-shoot and the K-Gun

By the late 1980s the next offensive answer was simply *more receivers*. If the 46 and the blitz-heavy defenses of the decade were built to win with numbers near the line, an offense could take the numbers away by spreading out.

The **run-and-shoot** went furthest: four wide receivers, one back, no tight end, and routes chosen after the snap from what the coverage did. Mouse Davis and June Jones, its two great teachers, brought it to the NFL, and between 1989 and the mid-1990s it was the base offense of the Detroit Lions, the Houston Oilers and the Atlanta Falcons. Warren Moon led the league in passing yards for Houston in 1990 and 1991. The defenses' answer was the one Ryan always preached: hit the quarterback. Four receivers meant only five or six blockers, and blitzes that brought more rushers than that forced the quarterback to throw before the routes had sorted themselves out. Ryan, by then Houston's defensive coordinator, called his own team's offense the “chuck and duck,” and on January 2, 1994 he threw a punch at its coordinator, Kevin Gilbride, on the sideline during a nationally televised game.¹⁸ The language and the reads of the run-and-shoot are in Offensive Systems.

Buffalo's version was more durable. The **K-Gun** was a no-huddle offense run with three wide receivers, one tight end and one back (11 personnel), with quarterback Jim Kelly calling the plays at the line. Its weapon was the clock between plays. A defense matches the offense's personnel by substituting when the offense does; when the offense doesn't substitute and doesn't huddle, the defense has no time to change, and is stuck with whatever it had on the field when the drive started (the history, including the 1989 attempt to penalize the no-huddle, is in In-Game Adjustments and Tempo). The 1990 Bills went 13–3, scored a league-high 428 points and beat the Raiders 51–3 in the AFC Championship Game. They went on to play in four straight Super Bowls, and lost all four.

¹⁶“Super Bowl XXII,” Wikipedia (Washington 42, Denver 10, January 31, 1988; 35 points in the second quarter; Timmy Smith 204 rushing yards, a Super Bowl record; Ricky Sanders 193 receiving yards): https://en.wikipedia.org/wiki/Super_Bowl_XXII ; “Doug Williams (quarterback),” Wikipedia (18 of 29 for 340 yards and four touchdowns, all in the second quarter; “the first African-American quarterback to both win a Super Bowl and be named its MVP”): [https://en.wikipedia.org/wiki/Doug_Williams_\(quarterback\)](https://en.wikipedia.org/wiki/Doug_Williams_(quarterback)) .

¹⁷“Body Bag Game,” Wikipedia (November 12, 1990; Eagles 28, Washington 14; nine Washington players injured, including quarterbacks Jeff Rutledge and Stan Humphries): https://en.wikipedia.org/wiki/Body_Bag_Game ; “1990–91 NFL playoffs,” Wikipedia (Washington 20, Philadelphia 6 in the wild-card round): https://en.wikipedia.org/wiki/1990%E2%80%9391_NFL_playoffs .

¹⁸“Buddy Ryan,” Wikipedia (as Houston's defensive coordinator, Ryan called the Oilers' offense the “chuck and duck” and “attempted to punch” offensive coordinator Kevin Gilbride on the sideline on January 2, 1994, against the Jets): https://en.wikipedia.org/wiki/Buddy_Ryan . The run-and-shoot's NFL teams and Warren Moon's passing titles are sourced in Offensive Systems.

The game: Giants 20, Bills 19, at Tampa on January 27, 1991, decided when Scott Norwood's 47-yard field goal try went wide right with eight seconds left. The Giants were heavy underdogs to the K-Gun, which had scored 95 points in two playoff games.¹⁹

What to notice: both of the Giants' game plans answer the no-huddle. On defense, Bill Belichick played extra defensive backs to take away Buffalo's passing game and told his players, who had been the league's best run defense, that they would win if Thurman Thomas ran for more than 100 yards. Thomas ran for 135 and the Giants won. On offense, Parcells kept the K-Gun on the sideline: the Giants held the ball for 40 minutes and 33 seconds, still the Super Bowl record. You can't use tempo while you're standing on the sideline. Belichick's defensive plan is now in the Pro Football Hall of Fame.

Safe pressure: the zone blitz meets the spread

The spread offenses beat man-coverage blitzes with **hot throws**: when more rushers come than the protection can block, the quarterback throws immediately to the receiver the blitzer left, and with four or five receivers spread wide, someone is always open for a moment. The 1990s defensive answer went back to Arnsperger's idea of safe pressure: rush five, but from places the protection doesn't expect, and drop a lineman into the throwing lane the quarterback expects to be open. The zone blitz gave a defense the pressure of a blitz without the man coverage that hot throws punish.

Who invented it is disputed. Some histories credit Arnsperger's Miami defenses, in the 1970s or in his second stint in the 1980s with the "Killer B's"; others credit Dick LeBeau, Cincinnati's coordinator from 1984, who studied Arnsperger and turned the idea into a system. This book gives Arnsperger the idea and LeBeau the system (the case is laid out in Zone Blitzes). It became famous in Pittsburgh, where Dom Capers and LeBeau built "Blitzburgh": the 1994 Steelers led the league with 55 sacks. Capers then took it to the expansion Carolina Panthers in 1995, and in their second season those Panthers went 12–4 and reached the NFC Championship Game.²⁰

Notice the shape of the exchange. The 1978 rules made passing easier; offenses spread out to pass more; defenses answered by pressuring the quarterback in ways he couldn't read before the snap. It is the same exchange you'll see again with the spread offenses and two-high defenses of The Modern Era.

1994: the league tilts the board again

Look back at the points panel of Figure 2. By 1992 and 1993 scoring had fallen to 18.7 points a game, the lowest since 1977, as the zone blitz, better pass rushers and faster defenses caught up. The owners responded as they had in 1978: in March 1994 they adopted a package of changes "to increase offensive production." It modified line play, the chucking rules and the roughing-the-passer rule, adopted the neutral zone infraction (a defender who enters the neutral zone and draws a flinch stops the play dead), and made two changes that are easy to see on any broadcast.²¹

The two-point conversion. After a touchdown, a team may now try to score two points by running or passing the ball into the end zone from the 2-yard line, instead of kicking for one. College football had used it since 1958 and the American Football League all through the 1960s; the NFL took 25 years after the merger to adopt it. The first one in NFL history was scored by Cleveland's punter and holder Tom Tupa, running in a faked kick in the opening week of 1994.²²

¹⁹"Super Bowl XXV," Wikipedia (Giants 20, Bills 19, January 27, 1991; Buffalo's 13–3 record, 428 points and 51–3 AFC Championship win; the K-Gun's 95 points in two playoff games; Belichick's plan to play extra defensive backs and his message that the Giants would win if Thomas ran for more than 100 yards, from David Halberstam's *The Education of a Coach*; Thomas's 135 rushing yards; the Giants' Super Bowl record 40:33 time of possession; Belichick's game plan in the Hall of Fame): https://en.wikipedia.org/wiki/Super_Bowl_XXV

²⁰"Zone blitz," Wikipedia (attributes the zone blitz to Arnsperger's Dolphins): https://en.wikipedia.org/wiki/Zone_blitz ; "Dick LeBeau," Wikipedia (Bengals defensive coordinator from 1984): https://en.wikipedia.org/wiki/Dick_LeBeau ; the Steelers' 55 sacks in 1994 and the "Blitzburgh" nickname are sourced in Zone Blitzes; "1996 Carolina Panthers season," Wikipedia (12–4, NFC West champions, beat Dallas 26–17, lost the NFC Championship Game 30–13 at Green Bay; Dom Capers head coach): https://en.wikipedia.org/wiki/1996_Carolina_Panthers_season

²¹Pro Football Hall of Fame, Chronology of Football, 1994 ("In a move to increase offensive production, NFL clubs ... adopted a package of changes, including modifications in line play, chucking rules and the roughing-the-passer rule, plus the adoption of the two-point conversion and moving the spot of the kickoff back to the 30-yard line," March 22): <https://www.profootballhof.com/football-history/football-history/1980-1999/1994> ; "1994 NFL season," Wikipedia (the neutral zone infraction adopted): https://en.wikipedia.org/wiki/1994_NFL_season . The 1992 and 1993 scoring averages are from the Pro Football Reference table cited with the chart.

²²"Two-point conversion," Wikipedia (the AFL used it from 1960 to 1969; the NFL adopted it in 1994; "Tom Tupa scored the first two-point conversion in NFL history, running in a faked one-point attempt for the Cleveland Browns in a game against the Cincinnati Bengals in the first week" of the 1994 season): https://en.wikipedia.org/wiki/Two-point_conversion . College football adopted the two-point try in 1958.

The decision is a small expected-points problem, the kind Expected Points teaches. Kicking was worth almost exactly one point: from 1999 to 2014 (the seasons in nflverse data before the extra point was moved back in 2015), NFL kickers made 98.8 percent of extra points. Going for two succeeded 46.9 percent of the time, which is worth $2 \times 0.469 \approx 0.94$ points.²³ So on average the kick was slightly better, and teams went for two only a few times a season each. The two-point try is a *situational* weapon: late in a game, when the score makes two points worth more than twice one point, the arithmetic flips, and that is what the coaches' two-point chart in Clock and Game Management encodes.

Kickoffs from the 30. The kicking spot moved back five yards, from the 35 to the 30. A kickoff from five yards further back lands five yards shorter, so fewer kicks reach the end zone and more are returned: in 1993 only 68 percent of kickoffs were returned; in 1994, 88 percent. More returns meant better starting field position for offenses, which is what the package was for. (The kickoff spot went back to the 35 in 2011, and the kickoff has since been rebuilt entirely; see Kickoffs and Returns.)²⁴

Scoring rose to 20.3 points a game in 1994 and 21.5 in 1995. The 1978 pattern again: the rules open a gap, and offenses move in.

Free agency and the salary cap: changing how teams are built

The last two changes of the era weren't about the field at all, and they may have changed the game more than any rule about hands. Until 1993, an NFL player whose contract ended could not simply sign with another team. Through the 1980s the league's rules made movement rare, and from 1989 to 1992 a system called **Plan B** let each team protect 37 players and leave only the rest free to sign elsewhere. Players sued. In 1992 a federal jury in a case brought by eight players, led by the Jets' Freeman McNeil, found that Plan B violated the antitrust laws, and in January 1993 the league and the players settled the remaining suits with a seven-year agreement that rebuilt the system.²⁵

Free agency (1993)

Free agency (1993) is *the system, in force since March 1, 1993, that lets a veteran whose contract has expired sign with any team*. Today a player with four accrued seasons becomes an unrestricted free agent; younger players are restricted, meaning their old team can match an offer. The first great free-agent signing showed what it could do. Reggie White, the best defensive end of his era, left Philadelphia in 1993 for a four-year, \$17 million contract with Green Bay, then the third-largest in the league; in 1996 he set a Super Bowl record with three sacks as the Packers won Super Bowl XXXI.²⁶

The salary cap (1994)

In exchange, the owners got the **salary cap**: *a hard limit, the same for every team, on what a club may count against its player payroll in a season, set each year from a share of league revenue*. It began in 1994 at \$34.6 million per team, higher than the expected \$32 million because a new television deal with a new network, Fox, raised league revenue. It is a *hard* cap: no team may exceed it for any reason, and signing bonuses are spread across the years of a contract so they can't be hidden.²⁷

! Why it exists

The two came as a pair, and each answers the other's problem. Free agency gave players the right to sell their services to the highest bidder; the cap kept the richest owners from simply buying every good player, which the league feared would end competitive balance. The trade the players made was a fixed share of revenue in return for movement; the trade the teams made was freedom to spend in return for a ceiling that made every contract a choice against every other.

How the cap changed football strategy. You will see three of its effects in almost every chapter of this book.

²³Course calculation from nflverse play-by-play (`load_pbp`), regular seasons 1999–2014: 18,592 extra-point kicks, 98.8% good; 1,041 two-point attempts, 46.9% successful (`two_point_conv_result == "success"`). The extra-point kick moved from the 2-yard line to the 15 in 2015.

²⁴"Kickoff (gridiron football)," Wikipedia ("In 1994, after only 68% of kicks were returned the prior year, the kickoff was moved to the 30-yard line. This increased returns to 88% during the 1994 season"; the spot moved back to the 35 in 2011): [https://en.wikipedia.org/wiki/Kickoff_\(gridiron_football\)](https://en.wikipedia.org/wiki/Kickoff_(gridiron_football))

²⁵"Free agent," Wikipedia, section on the NFL ("Plan B free agency ... active in the National Football League in February 1989 to 1992 ... permitted all teams in the NFL to preserve limited rights of no more than 37 total players"; "Eight players sued ... In 1992, a jury found that Plan B violated antitrust laws"; "The NFL's current free agency system was introduced on March 1, 1993"; unrestricted free agents today have four or more accrued seasons): https://en.wikipedia.org/wiki/Free_agent ; Pro Football Hall of Fame, Chronology of Football, 1989 (229 Plan B free agents signed with new teams) and 1993 (January 6: settlement and "a seven-year deal that included a new player system"): <https://www.profootballhof.com/football-history/football-history/1980-1999/1993> .

²⁶"Reggie White," Wikipedia (signed with Green Bay in 1993 for four years and \$17 million, then the third-highest-paid player behind John Elway and Dan Marino; three sacks in Super Bowl XXXI, a single-game Super Bowl record): https://en.wikipedia.org/wiki/Reggie_White

²⁷"Salary cap," Wikipedia, NFL section ("The cap was first introduced for the 1994 season and was expected to be \$32 million, but an unexpectedly high bid from Fox and other networks increased the cap to \$34.6 million"; a hard cap; signing bonuses prorated over the contract): https://en.wikipedia.org/wiki/Salary_cap ; Pro Football Hall of Fame, Chronology of Football, 1993 (Fox's new NFC package, December 18), cited above.

1. **Positions got prices.** Under a cap, paying one player means not paying another, so teams had to decide which positions are worth the most. The answers that emerged in the 1990s still hold: quarterback first, then the players who decide whether the quarterback has time, the pass rusher and the blind-side tackle. The cap is what turned Lawrence Taylor's effect into a salary structure. Building a Roster follows that logic to today's contracts.
2. **Rosters churned, so systems had to travel.** With players moving every spring, a team could no longer assume its starters had spent five years learning its playbook. Offenses and defenses had to be teachable to a newcomer in one off-season, which rewarded concept-based languages (one word for a whole pattern, the Erhardt-Perkins idea) and coaches who could install fast. Coaches moved too, and took their systems with them; the coaching trees of Part 12 are a product of the same mobility.
3. **Depth got thinner, and cheap players got valuable.** A team at the cap can't stockpile veterans, so the draft, where players are cheap for their first contract, became the most valuable source of talent, and specialists who play only in certain packages (a third-down pass rusher, a nickel corner) are bought cheaply rather than kept on the roster as insurance.

The cap also invites accounting creativity. In the early 2000s the league found that Denver had deferred money owed to John Elway and Terrell Davis outside the cap between 1996 and 1998, years in which the Broncos won two Super Bowls.

Where the era landed: 1998 and 1999

By the end of the 1990s, each of the era's big ideas had a winner. In 1998 the Denver Broncos went 14–2 behind Mike Shanahan's offense, which married Walsh's play-action passing to the outside-zone running Alex Gibbs had brought to Denver in 1995 (the story is in *Zone Running*). Terrell Davis ran for 2,008 yards, and John Elway, in his last game, was the MVP of a 34–19 win over Atlanta in Super Bowl XXXIII.²⁸ The same season the Minnesota Vikings, with a rookie receiver named Randy Moss, scored 556 points, then the most in league history.²⁹ And in 1999 the St. Louis Rams, running Mike Martz's Coryell-descended offense, the "Greatest Show on Turf," beat the Tennessee Titans 23–16 in Super Bowl XXXIV.³⁰

Three champions, three lines of descent from 1978: Walsh's timing game, Coryell's vertical game, and the run game that the one-back offenses had rebuilt to make both work. Meanwhile the defenses had their nickel and dime packages, their 3-4 edge rushers and their zone blitzes, and were waiting for the next rule change. The next round of the story, the spread offenses that came up from college and the defenses built to stop them, is in *The College Laboratory and Dynasties and Counterpunches*.

The whole era on one page

Figure 8 puts the chapter's moves and counter-moves side by side, year by year. Read across a row to see what was happening at the same moment, and down a column to follow one side's line of thought. (Entries not sourced earlier in the chapter come from the Hall of Fame's year-by-year chronology and the other notes listed here.)³¹

²⁸"1998 Denver Broncos season," Wikipedia (14–2; Terrell Davis 2,008 rushing yards; Super Bowl XXXIII, 34–19 over Atlanta, Elway MVP in his last game; the 2001–2005 investigation into deferred payments to Elway and Davis between 1996 and 1998): https://en.wikipedia.org/wiki/1998_Denver_Broncos_season

²⁹"1998 Minnesota Vikings season," Wikipedia (an NFL-record 556 points; rookie Randy Moss): https://en.wikipedia.org/wiki/1998_Minnesota_Vikings_season

³⁰"Super Bowl XXV," cited above, notes Super Bowl XXXIV (St. Louis 23, Tennessee 16) as the only other Super Bowl without a turnover; "Super Bowl XXXIV," Wikipedia: https://en.wikipedia.org/wiki/Super_Bowl_XXXIV. Martz's Coryell lineage: see the note on Coryell.

³¹Pro Football Hall of Fame, *Chronology of Football, 1980–1999* (the 57-day strike of 1982 and the nine-game season; limited instant replay adopted in 1986 and dropped after 1991; the 24-day strike of 1987 and the replacement games; 229 Plan B free agents signed in 1989): <https://www.profootballhof.com/football-history/football-history/1980-1999/1986>; "Instant replay in American football," Wikipedia (1986–1991; the current system with coaches' challenges from 1999): https://en.wikipedia.org/wiki/Instant_replay_in_American_football; "Dan Marino," Wikipedia (5,084 passing yards in 1984, a record until 2011): https://en.wikipedia.org/wiki/Dan_Marino; Fairbanks's and Phillips's 3-4s are sourced in *Odd and Hybrid Fronts*, and the run-and-shoot teams in *Offensive Systems*.

Offense		Rules and business	Defense
1972			Arnsparger's "53": Matheson rushes or drops; Miami goes 17-0
1974		Downfield contact limited (the "Isaac Curtis rule"); kickoffs move back to the 35	Fairbanks's Patriots and Phillips's Oilers make the 3-4 a base defense
1977		One chuck per receiver; head slap banned. Scoring hits 17.2 points a team-game	
1978	Coryell takes over the Chargers	Mel Blount rule; open-hand pass blocking; 16 games; a seventh official	
1979	Walsh hired by the 49ers; Fouts passes for 4,082 yards	"In the grasp": the whistle protects the passer	
1981	Gibbs to Washington: one-back, H-back, counter trey		Lawrence Taylor drafted second overall
1982		57-day players' strike; a nine-game season	
1984	Marino passes for 5,084 yards		LeBeau becomes Cincinnati's coordinator; zone blitz grows
1985			Ryan's 46: Bears allow the fewest points, win Super Bowl XX 46-10
1986		Limited instant replay (through 1991)	Taylor: 20.5 sacks, league MVP
1987	Washington scores 35 in one quarter of Super Bowl XXII	24-day strike; replacement games	
1989		Plan B: limited free agency	
1990	Run-and-shoot in Detroit, Houston, Atlanta; Bills' K-Gun scores 428		Belichick's plan holds the K-Gun to 19 in Super Bowl XXV
1993		Free agency begins March 1; Reggie White signs with Green Bay	
1994		Salary cap (\$34.6 million); two-point conversion; kickoffs back to the 30	"Blitzburgh": Steelers lead the NFL with 55 sacks
1995	Shanahan and Alex Gibbs bring outside zone to Denver		Capers takes the zone blitz to expansion Carolina
1998	Terrell Davis runs for 2,008; Vikings score a record 556		
1999	Rams' "Greatest Show on Turf" wins Super Bowl XXXIV	Replay returns, with coaches' challenges	

Figure 8: The arms race, 1972 to 1999. Each row is a year with an important change. Read across for what each side, and the league office, was doing at the same time; read down a column to follow one line of thought. The 1978 row is outlined.

Watch for it on Sunday

💡 Watch for it: classic film

Find a full game from the 1980s or early 1990s (NFL Films and the league's own YouTube channel post many; Super Bowls XX, XXII and XXV are easy to find) and chart the first quarter with what you now know.

- **Personnel and formation.** Count the backs and tight ends on every snap. How often is it the pro set or the I with two backs? When does a third receiver appear, and does the defense answer with a fifth defensive back? Is anyone lined up as an H-back, a step off the line beside the tackle?
- **The front.** Three linemen with hands down and two standing at the ends (a 3-4), or four down (a 4-3)? If you see the center and both guards covered and a safety walked up into the box, you're looking at the 46.
- **The best pass rusher.** Find him before the snap. Which side is he on, and what does the offense do about it: an H-back next to his tackle, a back staying in, a tight end chipping him?
- **Contact on receivers.** Watch a receiver past five yards. Does anyone touch him before the ball is thrown? (After 1978, a flag if they do, but watch how much the officials let go in the 1980s.)
- **The clock between plays.** Does the offense huddle? If it doesn't, does the defense manage to substitute?
- **Then flip to a 2025 game** and run the same list. Note what disappeared (the fullback, the 46, the huddle on most plays) and what survived (the slant-flat, the counter, the stand-up edge rusher, the zone blitz).

Predict the play

💡 Predict the play 1: where does the H-back go?

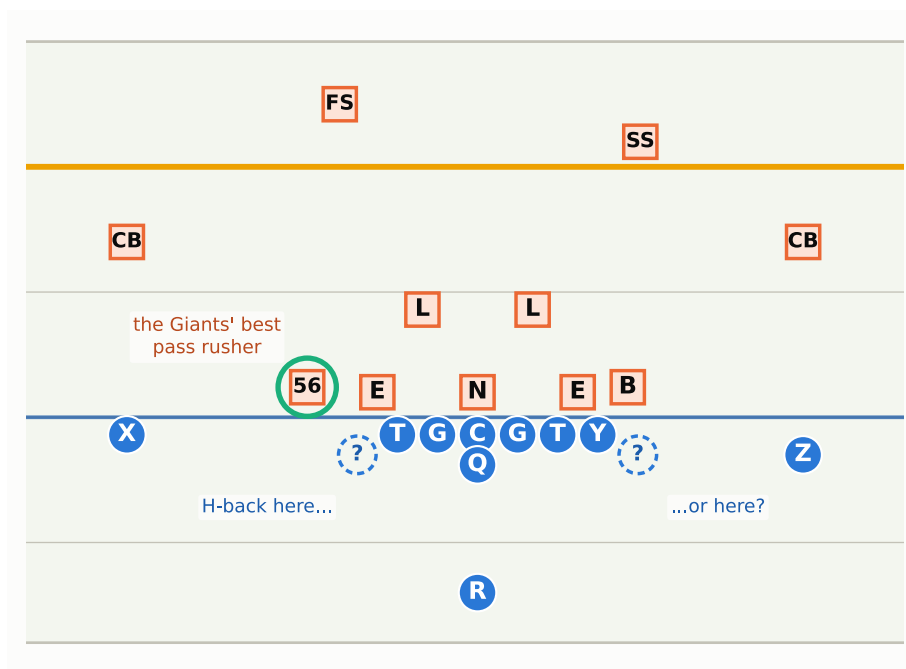


Figure 9: Predict the play 1. 1986: one back, a tight end (Y) on the right, and an H-back who could line up in either dashed spot. The Giants are in a 3-4; their best pass rusher, number 56 (ringed), stands at the end of the line on the offense's left.

It's 1986, and your right-handed quarterback is about to face the Giants. They are in their 3-4, and their best pass rusher, number 56, is standing at the end of the line on your left. You have one running back, a tight end on the right, and an H-back you can line up on either side. Where do you put the H-back, and what does he do at the snap?

Answer 1 is at the end of the chapter. Try it first.

💡 Predict the play 2: the K-Gun catches a base defense

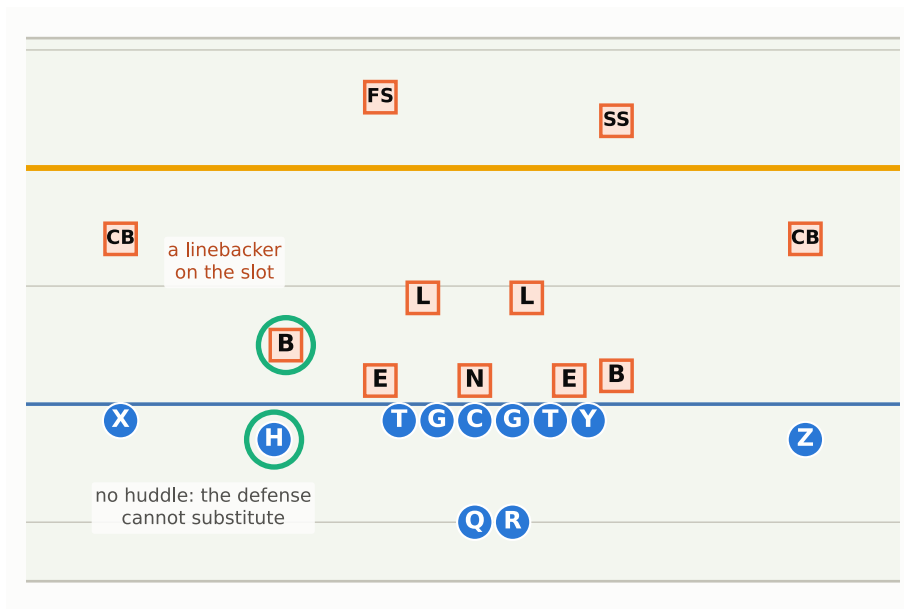


Figure 10: Predict the play 2. 1990: Buffalo in the shotgun with three wide receivers (X, H, Z), a tight end and a back, no huddle, against a base 3-4 that was on the field when the drive began. The slot receiver H and the linebacker over him (B) are ringed.

It's 1990. Buffalo gained 9 yards on first down and has hurried to the line without a huddle in the shotgun: three wide receivers, a tight end, one back. The defense had its base 3-4 on the field when the drive started, and it hasn't been able to change it. Where will Jim Kelly look, and what should the defense have done before the drive began?

Answer 2 is at the end of the chapter. Try it first.

💡 Predict the play 3: before the rule, and after

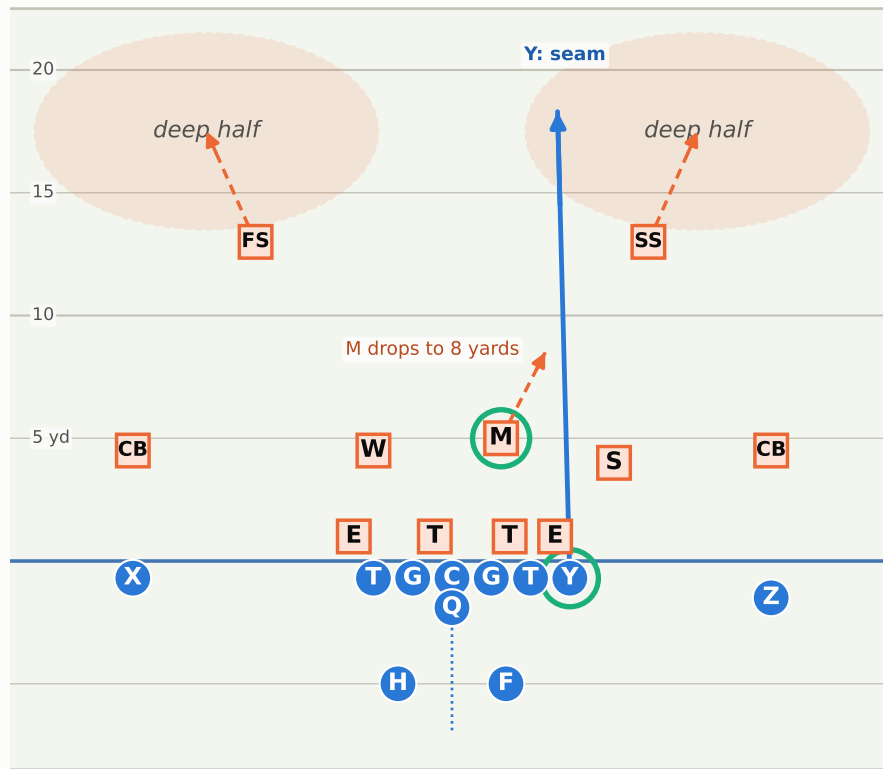


Figure 11: Predict the play 3. Cover 2 against a two-back pro set. The tight end (Y, ringed) runs up the seam between the two deep halves as the Mike (M, ringed) drops to about 8 yards in the middle.

The defense plays Cover 2: two deep safeties and five underneath. The Mike linebacker drops straight back to about 8 yards in the middle as the tight end, Y, releases up the seam between the two safeties. What does the Mike do to Y under the 1977 rules, and what can he do under the 1979 rules? And where is the ball going in 1979?

Answer 3 is at the end of the chapter. Try it first.

Takeaways

- **Two 1978 rules opened the passing game from both ends.** The Mel Blount rule stopped defenders from hitting receivers past five yards, which made timed routes possible; open-hand pass blocking gave the quarterback more time. Passes went from 40 percent of plays in 1977 to half of them by the early 1980s, and interceptions fell for twenty years.
- **The offense split into two answers.** Air Coryell stretched the field deep with protection and big receivers; Walsh's West Coast offense stretched it sideways with short, timed throws that worked like runs. Gibbs's one-back, two-tight-end offense, the run-and-shoot and the K-Gun built on both.
- **Every defensive answer answered something specific:** nickel and dime for the third receiver; the 53 and the 3-4 for disguising who rushes; Lawrence Taylor for slow protections; the 46 for two-back offenses; the zone blitz for hot throws. Each new answer created the next question.
- **The blind side made the left tackle a premium position,** because a right-handed quarterback can't see what comes from behind his left shoulder, and because the best rushers lined up there.
- **The league rewrote the rules again when defenses caught up:** the 1994 package added the two-point conversion and moved kickoffs back to the 30.
- **Free agency (1993) and the salary cap (1994) changed how teams are built:** positions got prices, systems had to be teachable fast, and cheap drafted players became the most valuable kind.

Connections

Builds on: From the Single Wing to the 1970s Scoring Drought (the problem the 1978 rules solved), Offensive Systems (the languages of Coryell, Walsh, Erhardt-Perkins and the run-and-shoot), Even Fronts (the 46) and Zone Blitzes (safe pressure).

Sets up: The College Laboratory (where the spread, the option and the RPO were being built while the NFL fought this war), Dynasties and Counterpunches (2000–2016), and the case studies of Part 12, starting with Walsh’s 49ers. For today’s contact rules, see Contact Rules and Penalties; for today’s cap economics, Building a Roster. Appendix E has the full rule-change timeline.

Go deeper

- Tim Layden, *Blood, Sweat and Chalk* (2010): chapters on Coryell, Walsh, Gibbs’s one-back and H-back, Ryan’s 46 and the zone blitz, with the coaches in their own words.
- Bill Walsh with Brian Billick and James Peterson, *Finding the Winning Edge* (1997): Walsh’s own account of his offense and how he built the 49ers.
- Michael Lewis, *The Blind Side: Evolution of a Game* (2006): Lawrence Taylor, the left tackle and the economics that followed.
- Chris B. Brown, *The Art of Smart Football* (2015): short essays on the arms race, including the zone blitz and the run-and-shoot.
- Pro Football Hall of Fame, “Chronology of Football” (profootballhof.com): the year-by-year record of rule changes used throughout this chapter.

Answers to Predict the play

Answer 1.

On the left, next to the left tackle. Number 56 is lined up over your quarterback’s blind side, so that is where you need an extra man. At the snap the H-back **chips** him: hits him as he comes off the edge, then releases into a short route if nothing else comes. That gives the left tackle help against a rusher he can’t block alone, and gives the quarterback a check-down when the H-back releases. This is the job Joe Gibbs created the position for.

On the right, the tight end is already in place to block. If 56 moved to the right side before the snap, as defenses learned to do, you would want your H-back to go with him, in motion across the formation. The rule is the one from Pass Protection: find the best rusher first and put help where he is, especially if he is on the blind side.

Answer 2.

At the slot receiver on the left (H). The only player to cover him in a base defense is a linebacker, the highlighted B, and a wide receiver against a linebacker in space is the matchup the K-Gun was built to create. Expect a quick throw to H (a slant, an option route that breaks away from the linebacker’s leverage) or a run against a defense with only four defensive backs spread across the field to cover four receivers.

The fix had to come *before* the drive, because once the offense stops huddling, the defense can only substitute when the offense does. A defense facing the K-Gun had to start its drives in nickel or dime and accept the trade: lighter against the run. That is exactly what Belichick’s Giants did in Super Bowl XXV, with extra defensive backs on the field and a promise to let Thurman Thomas run. And the Giants’ offense took away the other ingredient tempo needs: the ball. You can’t hurry the next snap if you aren’t on the field.

Answer 3.

1977: nobody touched Y at the line, so the Mike still has the one legal hit the 1977 rule allows, anywhere on the field (before 1974 he could have hit him as often as he liked). He meets Y at 8 yards, knocks him off his seam, and buys the safeties time to close the space between them. The route is ruined, and with it the timing.

1979: past five yards the Mike may not touch him. He has to turn and run with Y, a linebacker chasing a tight end up the field, or let him go and hope a safety gets there. Neither safety can, because each is responsible for his own half of the field, and the seam is exactly between them. The ball goes to Y at 18 to 20 yards, in the hole that every Cover 2 has in the deep middle.

The defense’s eventual answer was to make the Mike *run*: in Tampa 2 he sprints to the deep middle at the snap, turning Cover 2 into three deep in the middle of the field (Cover 2 and Tampa 2). That linebacker needed a sprinter’s speed, which is one more way the 1978 rules changed what a defensive player looks like.